Supplemental Material for
Mapping the interband Berry connection of an optical honeycomb lattice through its optical response

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EXPERIMENTAL SETUP

In our setup, ultracold gas mixtures of ^{40}K and ^{87}Rb are produced by laser cooling, co-trapped in a plugged spherical-quadrupole trap and then an optical trap. Forced evaporation of the ^{87}Rb atoms produces quantum degenerate gases of either ^{40}K, ^{87}Rb, or both elements simultaneously. In this paper, we produce a degenerate Fermi gas of around 8×10^4 ^{40}K atoms at a temperature of $\sim 0.2\,T/T_{\mathrm{F}}$ in a crossed optical dipole trap by evaporating away all ^{87}Rb atoms. T_F is the Fermi energy, which is around $h \times 10$ kHz. We note that the average of the local Fermi energy is around $h \times 6$ kHz, consistent with our observation that the third and higher bands are completely empty, and that the ground band is significantly more populated than the second band.

The optical honeycomb lattice is formed by three phase-coherent light beams at $\lambda = 1064\,\mathrm{nm}$, propagating at 120° with respect to each other in the x-y-plane perpendicular to gravity. The resulting potential is given by

$$V(\mathbf{r}) = -\frac{2}{9} V_0 \left(3 - \cos(\mathbf{G}_1 \cdot \mathbf{r}) - \cos(\mathbf{G}_2 \cdot \mathbf{r}) - \cos(\mathbf{G}_3 \cdot \mathbf{r})\right)$$

where $\mathbf{G}_1 = \mathbf{k}_2 - \mathbf{k}_3$ and $\mathbf{G}_2 = \mathbf{k}_3 - \mathbf{k}_1$ are primitive reciprocal lattice vectors, and $\mathbf{G}_3 = \mathbf{k}_1 - \mathbf{k}_2$. At the full lattice depth $V_0 = 8.95\,E_{\mathrm{R}}$ used in this work, the lattice beams and trapping beams creates an approximately harmonic potential with trap frequencies $\omega_{x,y,z} = 2\pi \times (69,\, 97,\, 353)\,\mathrm{Hz}$.

The data shown in the main text are taken with a total modulation time between 1.00 and 1.06 ms. The modulation amplitude of lattice shaking is ramped up in 150 µs, and then left at constant modulation amplitude. We enforce that the shaking ends with a non-moving lattice to avoid unwanted displacement in the final image. For linear shaking, this is done by choosing the initial phase of the modulation such that the lattice is stationary at the end. For circular shaking, since the speed of the lattice is constant at a given amplitude, we ramp down the modulation amplitude to zero in the last 150 µs instead. When using Eq. (2) in the main text to extract the value of interband Berry connection, we account for the ramp and the small steps in modulation time by using the pulse area for t_{PM}.

The position of the lattice is feedback stabilized by a phase lock setup [1] whenever the lattice depth is larger than $0.05E_R$. We confirm that the deviation from the programmed displacement is less than $a_{\mathrm{hc}}/10$ at any time during lattice shaking.

The quasimomentum resolution of our experiment is limited by the random walk of atom position as the atom scatters light from the imaging pulse during our absorption imaging sequence. The standard deviation of the distribution in final atom position after an 80 µs imaging pulse with intensity $0.15\,I_{\mathrm{sat}}$ is 3.38 µm, where I_{sat} is the saturation intensity. This deviation is comparable to the resolution of our imaging system, which is limited by the pixel size of the camera to be 3.7 µm. We convolve each averaged image with a one-pixel wide normalized Gaussian mask before further analysis.

We restrict our analysis of the dynamics of the system using Eq. (2) in the main text, which holds when the perturbation is close to the resonance. To confirm that a first-order perturbation theory treatment appropriately describes our measurement results, we perform two experiments at a representative modulation frequency $\omega_{\mathrm{PM}} = 2\pi \times 14.35\,\mathrm{kHz}$. We first fix $t_{\mathrm{PM}} = 1\,\mathrm{ms}$ and vary $0.08 \leq F/(E_R/a_{\mathrm{hc}}) \leq 0.43$. A log-log fit to D_{14} as a function of F gives scaling factors reasonably close to the expected quadratic scaling, see Fig. S1. In the second experiment, we shake at $F/(E_R/a_{\mathrm{hc}}) = 0.38$, while varying t_{PM} up to 3 ms. The measured excitation ratio deviates strongly from the expected quadratic scaling for both shaking strengths, especially at longer times, see Fig. S2. We emphasize that this deviation is not explained by two-level system dynamics, from which one expects Rabi-like oscillatory behavior. Instead, we attribute this to the dynamics of holes in the ground band [2], which is supported by the observation that at longer times the measured excitation fraction becomes non-zero even at off-resonant $\mathbf{q}$.

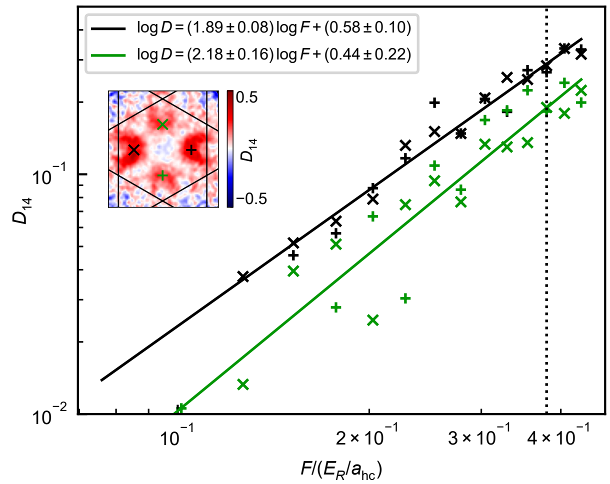

Fig. S1. Horizontal shaking at 14.35 kHz at different modulation strengths. Each set of data is taken from the two parity-symmetric positions marked in the inset. Solid lines are fits to linear functions. The black dashed line labels $F/(E_R/a_{\rm hc}) = 0.38$ used in the main text. Inset: an example run at $F/(E_R/a_{\rm hc}) = 0.38$ with quasimomenta markers.

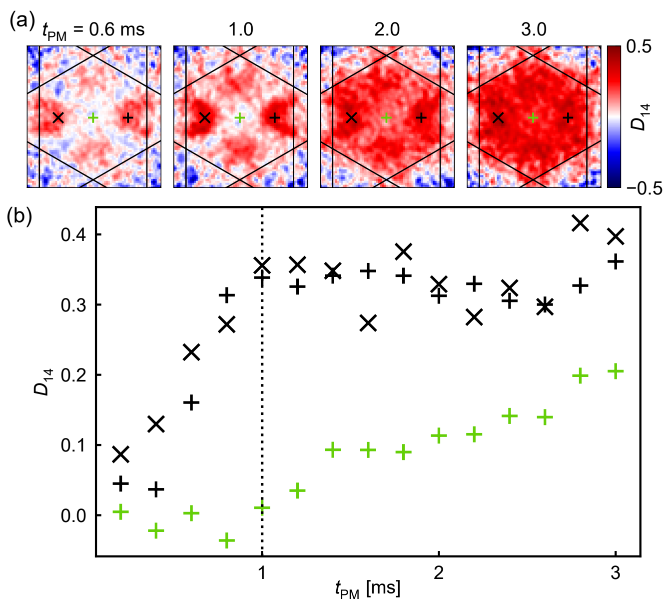

Fig. S2. Horizontal shaking at 14.35 kHz for varying durations. (a) D_{14} measurements for selected $t_{\rm PM}$ with quasimomenta markers for (b). (b) D_{14} as a function of $t_{\rm PM}$. Black dashed line labels $t_{\rm PM} = 1\,$ms used in the main text.

TIGHT BINDING MODEL FOR THE LOWEST TWO BANDS

In tight-binding model, the momentum space Hamiltonian for the honeycomb lattice is given by

$$H(\mathbf{q}) = -J \begin{pmatrix} 0 & g(\mathbf{q}) \\ g^*(\mathbf{q}) & 0 \end{pmatrix}, \tag{S1}$$

where $g(\mathbf{q}) = \sum_i e^{-i\mathbf{q}\cdot\mathbf{A}_i}$, J is the hopping strength, and $\mathbf{A}_i$'s are the vectors that connect one lattice A site to the nearest-neighbor B sites [see Fig. 1(c) of the main text]. For the honeycomb lattice studied in this paper, we have

$$\mathbf{A}_1 = a_{\rm hc}\begin{pmatrix} 1 \\ 0 \end{pmatrix},\ \mathbf{A}_2 = a_{\rm hc}\begin{pmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix},\ \mathbf{A}_3 = a_{\rm hc}\begin{pmatrix} -\frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}.$$

The interband Berry connection between the lower and upper bands can be expressed as

$$\mathbf{A}_{12}(\mathbf{q}) = \frac{1}{2|g(\mathbf{q})|^2}\sum_j C_j(\mathbf{q})\mathbf{A}_j, \tag{S2}$$

where $C_j(\mathbf{q}) = \sum_i \cos\left(\mathbf{q}\cdot(\mathbf{A}_i - \mathbf{A}_j)\right)$. The condition for the excitation rate to vanish is found by setting the matrix element $|\boldsymbol{\epsilon}(\theta_{\mathrm{PM}})\cdot\mathbf{A}_{12}(\mathbf{q})| = 0$, where $\mathbf{q} = q(\cos(\phi)\mathbf{e}_x + \sin(\phi)\mathbf{e}_y)$. To lowest order in q, we find $\cos(\theta_{\mathrm{PM}} + 2\phi) = 0$, giving four distinct solutions

$$\phi = \left(\frac{m}{2} + \frac{1}{4}\right)\pi - \frac{\theta_{\mathrm{PM}}}{2} \quad (m \in \mathbb{Z}). \tag{S3}$$

For large q, higher order terms modify the position of zeroes, but a numerical calculation based on plane wave expansion shows that the deviation is only around $\pi/30$ at the value of q we analyze, see Fig. 2(e) in the main text.

PLANE WAVE EXPANSION

The Hamiltonian for the stationary system $\hat{H}_0 = \hat{H}_{\mathrm{kin}} + V(\hat{\mathbf{r}})$ can be expressed in a plane wave basis. We keep plane waves with wave vectors up to $G_{\max} = 5\mathbf{G}$, where $\mathbf{G}$ is a primitive lattice vector. The Hamiltonian is diagonalized numerically and the eigenstates and eigenenergies at all quasimomenta are extracted. For Fig. 3 of the main text we write the interband Berry connection in terms of the velocity operator [3] and use

$$\langle n\mathbf{q}|\,\mathbf{v}\,|n'\mathbf{q}'\rangle = \frac{1}{m}\sum_{\mathbf{r},\mathbf{r}'}\langle n\mathbf{q}|\mathbf{r}\rangle\,\langle\mathbf{r}|\,\hat{\mathbf{p}}\,|\mathbf{r}'\rangle\,\langle\mathbf{r}'|n'\mathbf{q}'\rangle \tag{S4}$$

$$= \frac{\hbar}{m}\sum_{\mathbf{G}}(c_{\mathbf{G}}^{(n,\mathbf{q})})^*\,(\mathbf{q}+\mathbf{G})c_{\mathbf{G}}^{(n',\mathbf{q})}, \tag{S5}$$

$$= \frac{\hbar}{m}\sum_{\mathbf{G}}(c_{\mathbf{G}}^{(n,\mathbf{q})})^*\,\mathbf{G}\,c_{\mathbf{G}}^{(n',\mathbf{q})}, \tag{S6}$$

where $\langle\mathbf{r}|n\mathbf{q}\rangle = e^{i\mathbf{q}\cdot\mathbf{r}}\,\langle\mathbf{r}|u^{(n,\mathbf{q})}\rangle = e^{i\mathbf{q}\cdot\mathbf{r}}\sum_{\mathbf{G}} c_{\mathbf{G}}^{(n,\mathbf{q})}\,e^{i\mathbf{G}\cdot\mathbf{r}}$ and we have used Bloch state orthogonality in the last step.

INTERPRETATION OF DIFFRACTION IMAGING DATA

By abruptly turning off the lattice before imaging, we perform a projective measurement in the momentum state basis. As mentioned in the main text, the measured values of $D_{nn'}(\mathbf{q})$ are smaller than the excitation probabilities $P_{nn'}(\mathbf{q})$, because the momentum distribution of the $n = 1$ state is not entirely within the first Brillouin zone, and similarly the excited state in band n' may have nonzero amplitude in the first Brillouin zone. Quantitatively, let us consider two Bloch states (dropping $\mathbf{q}$ dependence for brevity)

$$|u_n\rangle = \sum_{\mathbf{G}} a_{\mathbf{G}}\,|\mathbf{G}\rangle\,,\ \ |u_{n'}\rangle = \sum_{\mathbf{G}} b_{\mathbf{G}}\,|\mathbf{G}\rangle\,,$$

where $|\mathbf{G}\rangle$ is the plane wave state with wave vector $\mathbf{G}$, and we make the phase choice so that $a_{\mathbf{0}}$ and $b_{\mathbf{0}}$ are real positive numbers for this calculation. Then the measured fractional change in column densities $(\rho_{\mathrm{ref}} - \rho_{\mathrm{shake}})/\rho_{\mathrm{ref}}$ is systematically smaller than the actual fractional decrease in the population in band n by a factor of $a_{\mathbf{0}}^2 - b_{\mathbf{0}}^2$.

In principle, since the excitation between $|u_n\rangle$ and $|u_{n'}\rangle$ is driven coherently, the relative phase between the two states should also affect our measurement results. We show that this coherent contribution is canceled out by our measurement protocol. More explicitly, we are interested in measuring the $|\mathbf{0}\rangle$ population of the state

$$|\Psi\rangle = A\,|u_n\rangle \pm Be^{i\varphi}\,|u_{n'}\rangle\,,$$

where A and B are positive real numbers, and $B = \sqrt{1-A^2}$. $+$ and $-$ signs correspond to shaking along θ_{PM} and $\theta_{\mathrm{PM}}+\pi$, respectively. In our sequence they correspond to the same polarization, but the modulations start at opposite phases. We have

$$|\langle\mathbf{0}|\Psi\rangle|^2 = a_{\mathbf{0}}^2\left(A^2 + r^2B^2 \pm 2rAB\cos\varphi\right), \tag{S7}$$

where $r = b_{\mathbf{0}}/a_{\mathbf{0}}$. The last term results from the coherence between $|u_n\rangle$ and $|u_{n'}\rangle$. Averaging the θ_{PM} and $\theta_{\mathrm{PM}}+\pi$ image then cancels the coherence term.

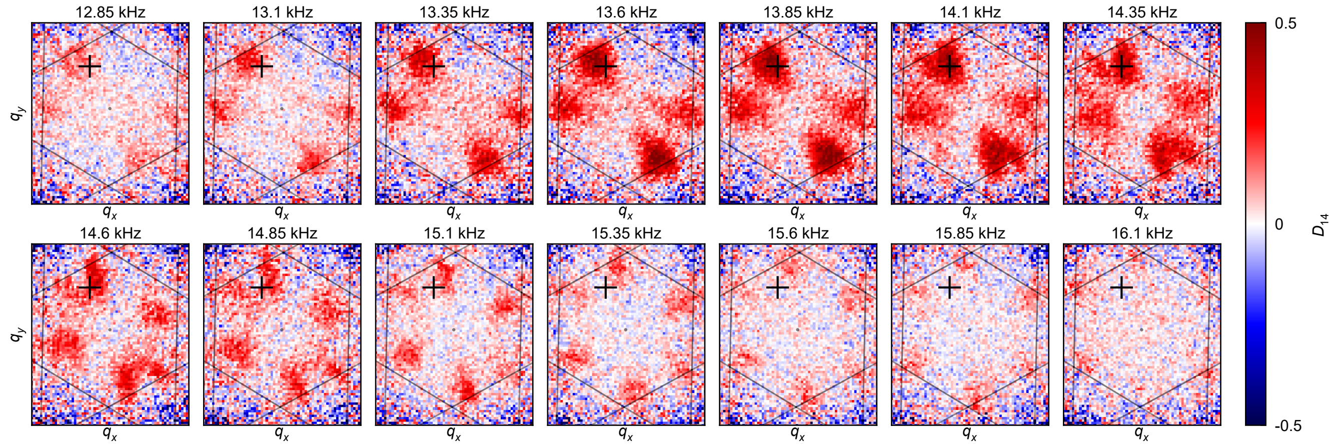

Fig. S3. The fractional depletion of atomic population, D_{14}, from diagonal ($\theta_{\rm PM} = 45°, 225°$) linear shaking. In the first Brillouin zone (hexagonal black outline), the atomic population depletes at different pixels, corresponding to different quasimomenta, when the shaking frequency (sub-figure labels) is resonant. The black cross marks the pixel analyzed in Fig. S4.

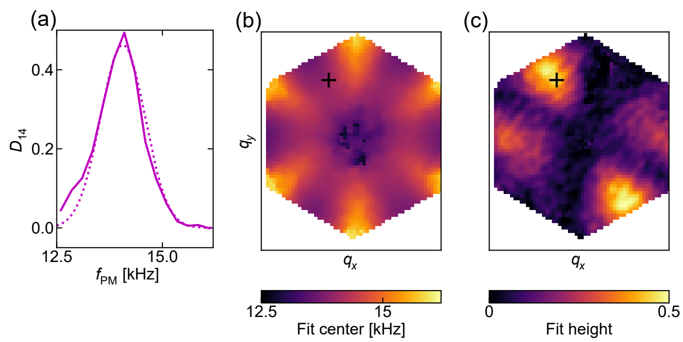

Fig. S4. Fitting the excitations from diagonal ($\theta_{\rm PM} = 45°$) linear shaking data. (a) Excitation spectrum at a single pixel marked in (b) and (c), and in Fig. S3. A Gaussian fit (dashed line) to the excitation spectrum (solid line) extracted from experiment reveals the resonant excitation frequency at one quasimomentum. (b) Performing such fits for each pixel in our images determines the resonant excitation frequency for transitions between bands across the entire Brillouin zone. (c) The extracted Gaussian fit height over the entire first Brillouin zone. The general structure is governed by the orientation and the strength of the interband Berry connection.

DETAILS ON DATA ANALYSIS FOR FIG. 2

The trial function used for fitting the angular integral of the annular region shown in Fig. 2(d)–(f) of the main text is given by

$$w(\phi) = A + \sum_{i=1}^{4} B_i \exp\left[\frac{\mathrm{mod}_{(-\pi,\pi]}[\phi - \phi_i]^2}{2\sigma_i^2}\right]$$

where ϕ is the polar angle shown in Fig. 2(d). $\mathrm{mod}_{(-\pi,\pi]}[\phi - \phi_i]$ means that we take the modulo of $\phi - \phi_i$ such that $-\pi < \phi - \phi_i \leq \pi$ for each ϕ_i. The angular positions of the transparency lines are then taken to be $\{\phi_1, \phi_2, \phi_3, \phi_4\}$.

EXTRACTION OF THE INTERBAND BERRY CONNECTION

After collecting data and reference images, we extract the center of mass of the unshaken reference images from atoms in a static lattice and we use it as the $\mathbf{q} = \Gamma$ position. The length of reciprocal lattice vectors are calibrated using diffraction images of ^{87}Rb Bose-Einstein condensates. Data of the same shaking direction and frequency are averaged over 2 - 6 runs.

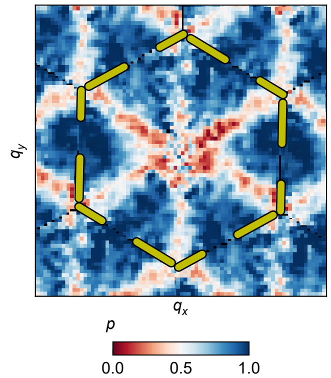

Fig. S5. p for $\mathbf{A}_{14}$ extracted from a four parameter fit. Yellow dashed lines mark the edge of the first Brillouin zone.

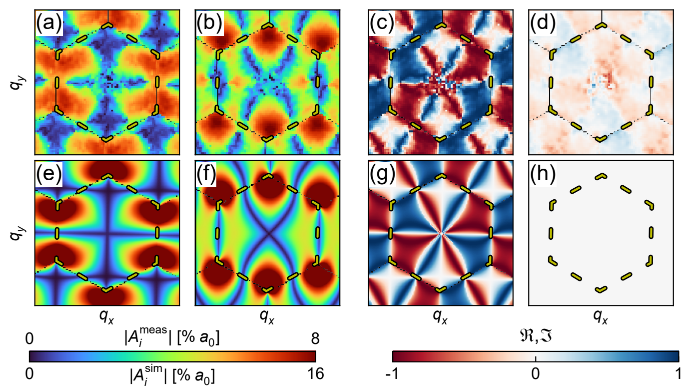

Fig. S6. Interband Berry connection $\mathbf{A}_{13}$ shown in a similar fashion as Fig. 3 of the main text.

We choose a fit function that incorporates all six shaking directions, and we fit the $D_{nn'}$ response vs. frequency to six Gaussians that share the same resonance frequency, but have different heights and widths, using a least squares fit. As an example, we show the averaged data in Fig. S3, which are fitted to get the results shown in Fig. S4. Fig. S4(a) shows an example pixel fit for a diagonal shaking case with an excitation from band 1 to 4. Fig. S4(b) and (c) show the fit center frequency, which can be used to extract energy band differences, and the fit heights, respectively, for the diagonal linear shaking over the entire Brillouin zone. The fitted peak heights for different shaking directions are used in another least squares fit that fits the data to the expected component of $\mathbf{A}$ according to Eq. 2 of the main text.

We examine whether the experimentally measured $D_{nn'}$ is explained by its relation with the interband Berry connection, as given in Eq. 2 of the main text, using a generalized set of fit parameters. Under the requirement that the probability of excitation must be real, the most general form of $P_{nn'}$ can be written as $P_{nn'}(\mathbf{q}) \propto \boldsymbol{\epsilon}^* \mathcal{A} \boldsymbol{\epsilon}$, where

$$\mathcal{A} = \begin{pmatrix} \mathcal{A}_{00} & \mathrm{Re}(\mathcal{A}_{01}) + i\,\mathrm{Im}(\mathcal{A}_{01}) \\ \mathrm{Re}(\mathcal{A}_{01}) - i\,\mathrm{Im}(\mathcal{A}_{01}) & \mathcal{A}_{11} \end{pmatrix} \tag{S8}$$

In general, $\mathrm{Re}(\mathcal{A}_{01})$ and $\mathrm{Im}(\mathcal{A}_{01})$ are independent of $\mathcal{A}_{00}$ and $\mathcal{A}_{11}$. Using Eq. 2 for analysis, as followed in the main text, is equivalent to taking $\mathcal{A} = \mathbf{A}\mathbf{A}^\dagger$, which contains only three independent parameters, since $\det(\mathcal{A}) = 0$. As explained in the main text, due to hole diffusion and finite momentum resolution, the measured $D_{nn'}$ at each pixel is slightly affected by the population depletion at neighboring pixels. As a result, we may find $\det(\mathcal{A}) > 0$ if the phase of the cross term is not constant in the vicinity of a pixel.

We analyze our data using a four parameter fit defined by Eq. S8, and define $p = \sqrt{1 - 4(\det(\mathcal{A})/(\mathrm{Tr}(\mathcal{A}))^2} = 1$ as a measure of the validity of the geometric interpretation. $p = 1$ indicates that $\mathcal{A} = \mathbf{A}\mathbf{A}^\dagger$. The result is shown in Fig. S5. A three parameter fit well describes the system in regions where $p \approx 1$. We find that $p \not\approx 1$ at quasimomenta where the amplitudes of the cross term are small.

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